

Enhancing NWP Precipitation Forecasting in the NE U.S.: Post-Processing with Deep Learning

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INTRODUCTION

- The majority of weather forecasts are created using Numerical Weather Prediction (NWP) models, which assimilate data from multiple sources and compute various physics-based equations to attempt to model the fluid dynamics of the atmosphere and oceans from the current state of the atmosphere to one timestep in the future.
- The conditions from that previously-computed timestep are then inputted to recursively derive the desired forecast. Small inaccuracies of each forecast multiply over time leading to increasing error as the number of timesteps in the future increases. Many of the processes that contribute to precipitation cannot be directly computed using physics equations and must be estimated, therefore making predicting precipitation and precipitation type a challenge even with advancements in NWP.
- Deep learning (DL) algorithms train input and output nodes in multiple layers through statistical techniques to identify patterns in data, leading to predictive analysis. These layers form a neural network which can represent non-linear relationships and handle high-dimensional data from multiple sources. Convolutional neural networks (CNNs) are a form of neural network that handles spatial data by including local patterns and identifying features and relationships within the spatial structure of input data.
- This research assesses the accuracy of CNN methods to predict precipitation by creating a CNN model using a U-Net architecture, which has historically been employed for image segmentation (Ronneberger et al., 2015). A U-Net is composed of several convolutional layers that extract information about a pixel and it's surrounding pixels, allowing the model to learn spatial relationships, which can be leveraged for weather forecasting.

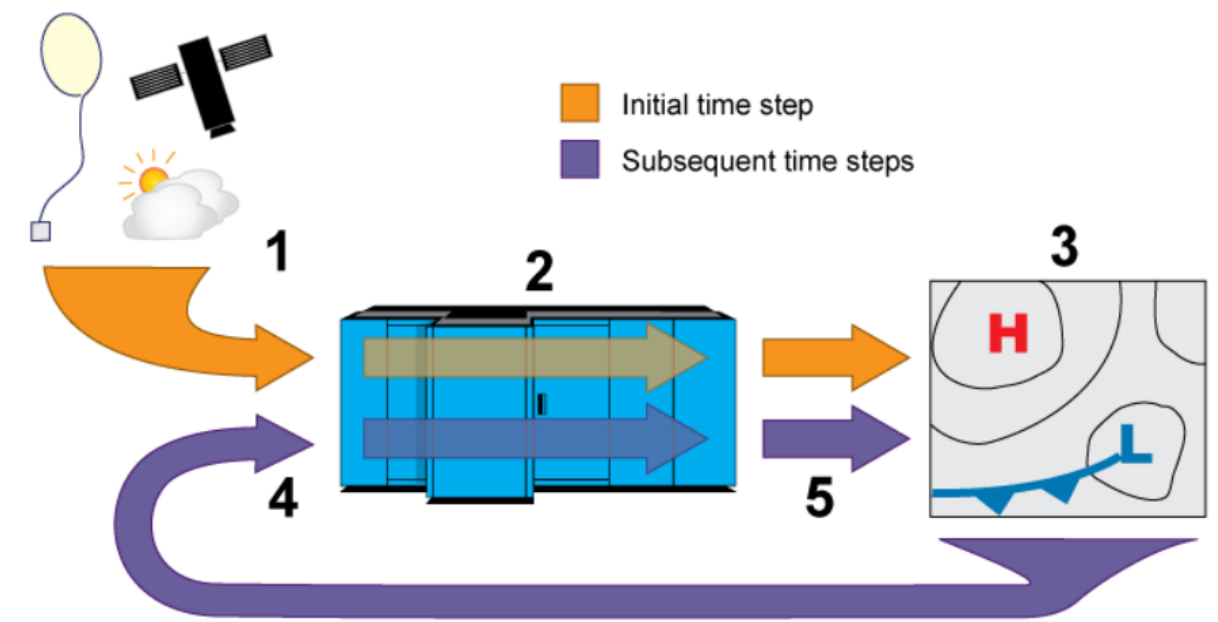


Figure 1. Simple Representation of NWP processing (NOAA 2023).

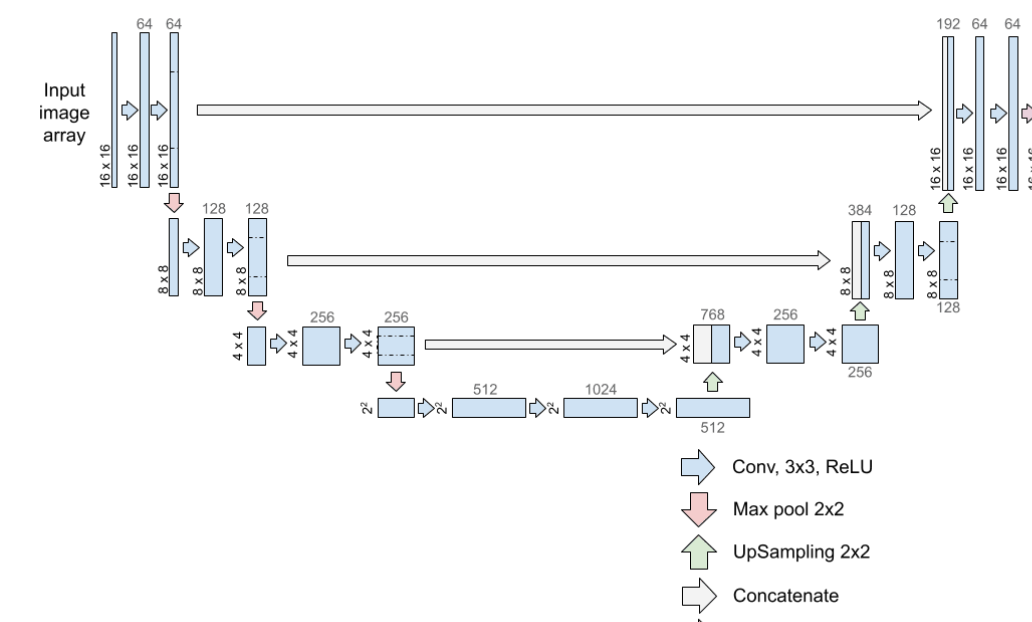


Figure 2. Representation of U-Net Architecture

MATERIALS

- Accessed through Google Earth Engine:
 - NWP Forecast: CFSV2: NCEP Climate Forecast System Version 2, 6-Hourly Products
 - Resolution: ~ 0.2 x 0.2 degree resolution
 - Initialized four times per day (0000, 0600, 1200, and 1800 UTC)
 - Observational Dataset: GSMap Operational: Global Satellite Mapping of Precipitation - V8
 - Resolution: ~ 0.1 x 0.1 degree resolution
 - Precipitation target: Hourly precipitation rate adjusted to rain gauge (averaged over the course of 6 hours to match NWP precipitation)
- Date Range:
 - One year (2020) of data was used to train the model due to computational constraints.
- The model was created using Google Collab and Keras TensorFlow API.

METHODS

- The study area includes New York State and portions of surrounding states.

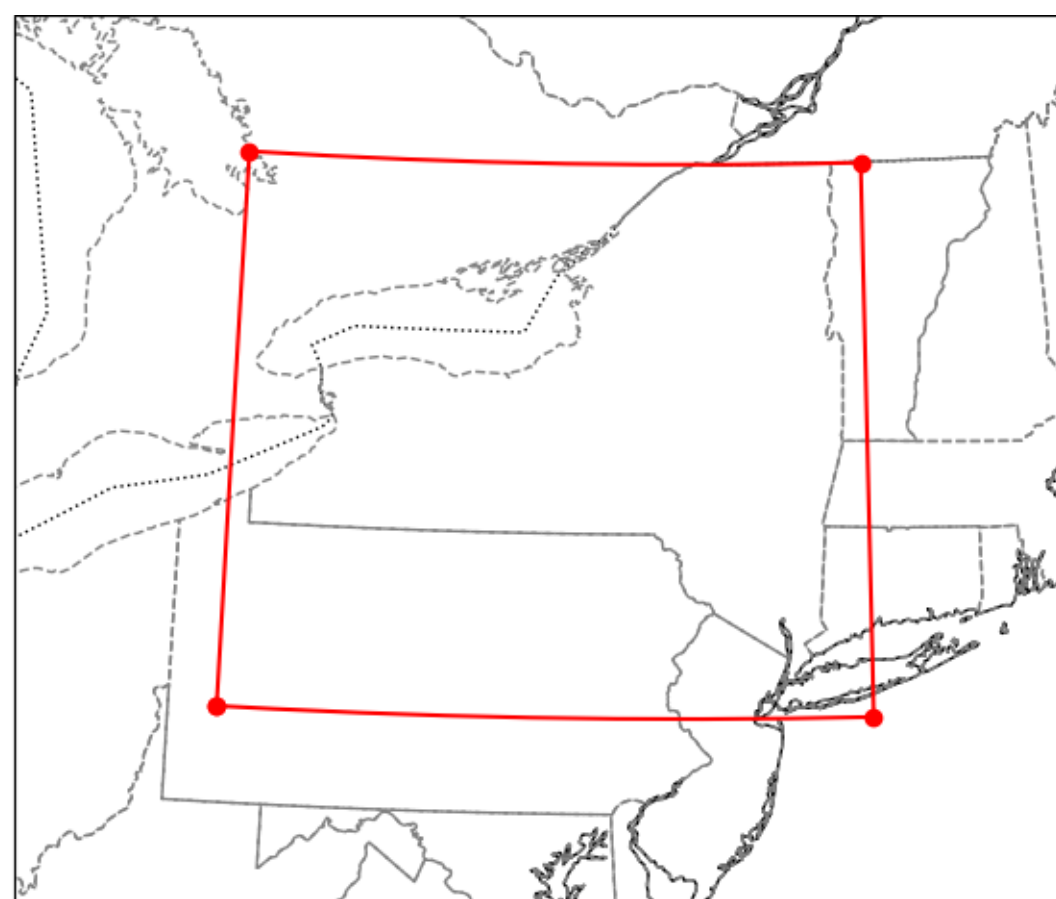


Figure 3. Study Area / Model Domain containing Latitudes: 40.5 to 45 and Longitudes: -80 to -73

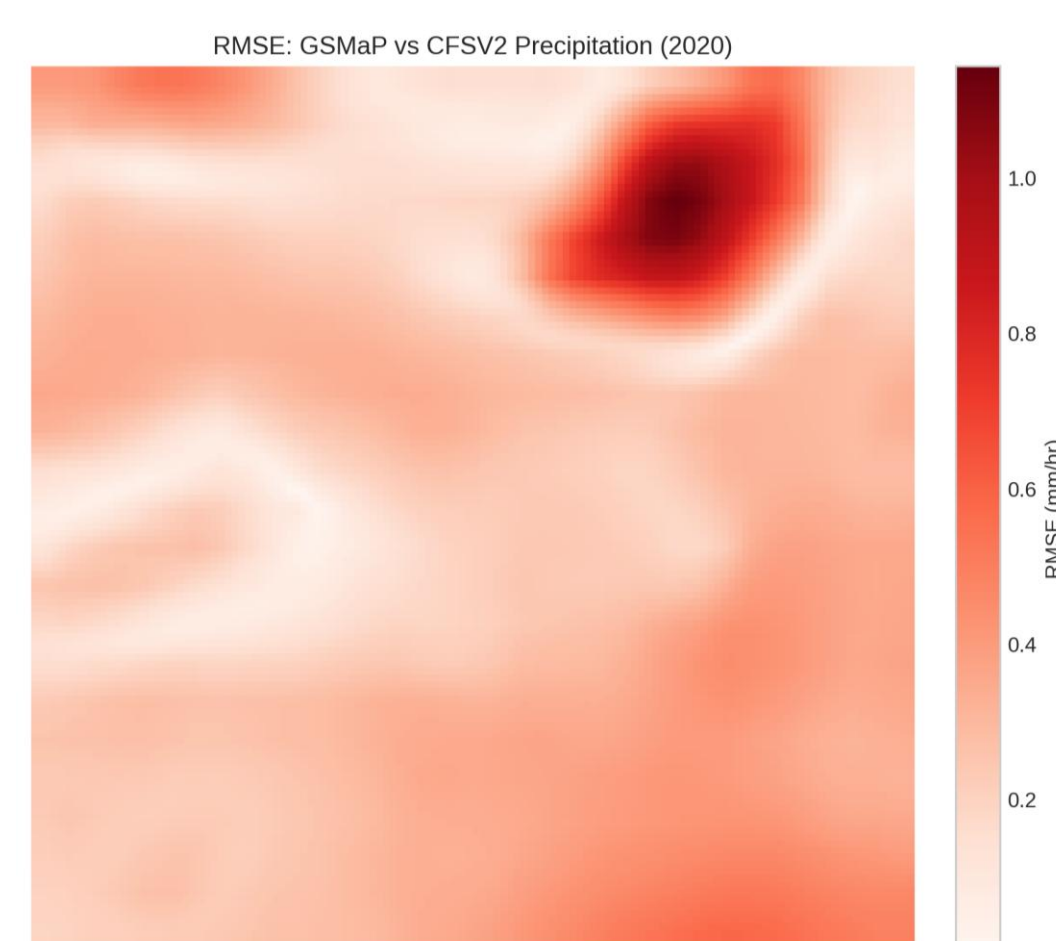


Figure 4. Root Mean Square Error of CFSV2 compared to GSMap precipitation in 2020.

- GSMap data was resampled to match the CFSV2 resolution.
- To select variables from the CFSV2 dataset, correlation coefficients with GSMap precipitation and surface variables were calculated using the Pearson and Spearman formulas.
- Notably, the correlation between the CFSV2 6-hour average precipitation rate and GSMap precipitation was very low, highlighting systematic or resampling errors.

- The RMSE between the two datasets can be seen in Figure 4., to the left.

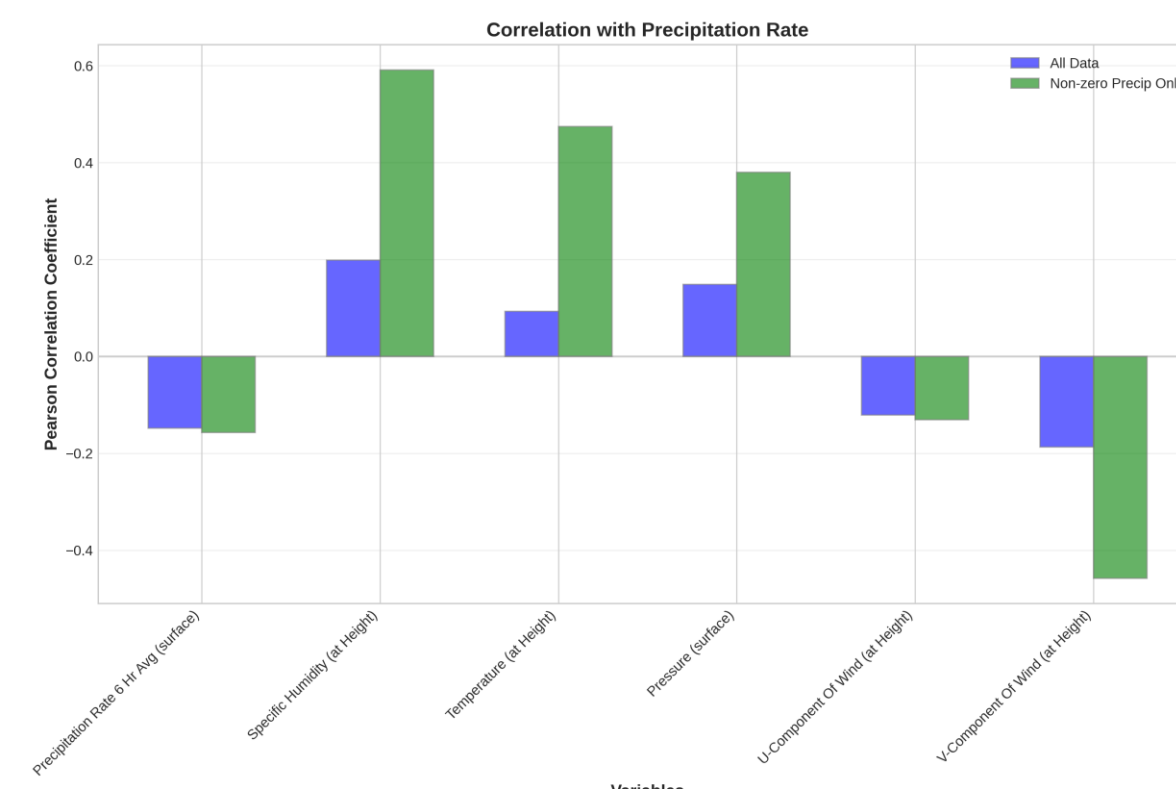


Figure 5. Pearson Correlation Coefficient of Selected Variables.

- There exists a major class imbalance in the distribution of precipitation as most hours and grid points do contain precipitation values.
- To more effectively visualize the relationship between selected feature variables and labeled precipitation, we create two scatter plots one with 5,000 sampled data points and the other with non-zero precipitation only.
- Below are the two scatter plots of the variables with the highest correlation to the labeled precipitation, specific humidity and temperature.



Figure 6. Pearson Correlation Coefficient of Selected Variables.

- Masked data was set to a default value, 0, as it generally corresponds to 0 precipitation in the Google Earth Engine environment. Precipitation values were also log-transformed to help deal with the class imbalance.
- Input features were normalized using the standard z-score normalization formula shown to the right.

$$z = \frac{x - \mu}{\sigma}$$

μ = Mean
 σ = Standard Deviation
- The total dataset was split into a training and validation based on time leading from past to present. The training dataset comprised ~80% of all hours. The validation dataset comprised ~20% of the remaining hours.
- The research utilizes pre-built layers through Keras including: input, 2D convolutional, max pooling, up sampling, and concatenate layers. Adam was chosen as the model optimizer, and best results came from a 0.0001 learning rate.
- Mean squared error was implemented as the loss function.

RESULTS

- A dataset from January 1, 2021, to February 28, 2021, was used to evaluate the performance of the model, of which there were twenty valid samples to compare.
- Figure 7., below, shows a time series of actual versus predicted mean precipitation across the entire domain, demonstrating high performance but a trend to underpredict moderate to large amounts of precipitation and overpredict trace amounts.

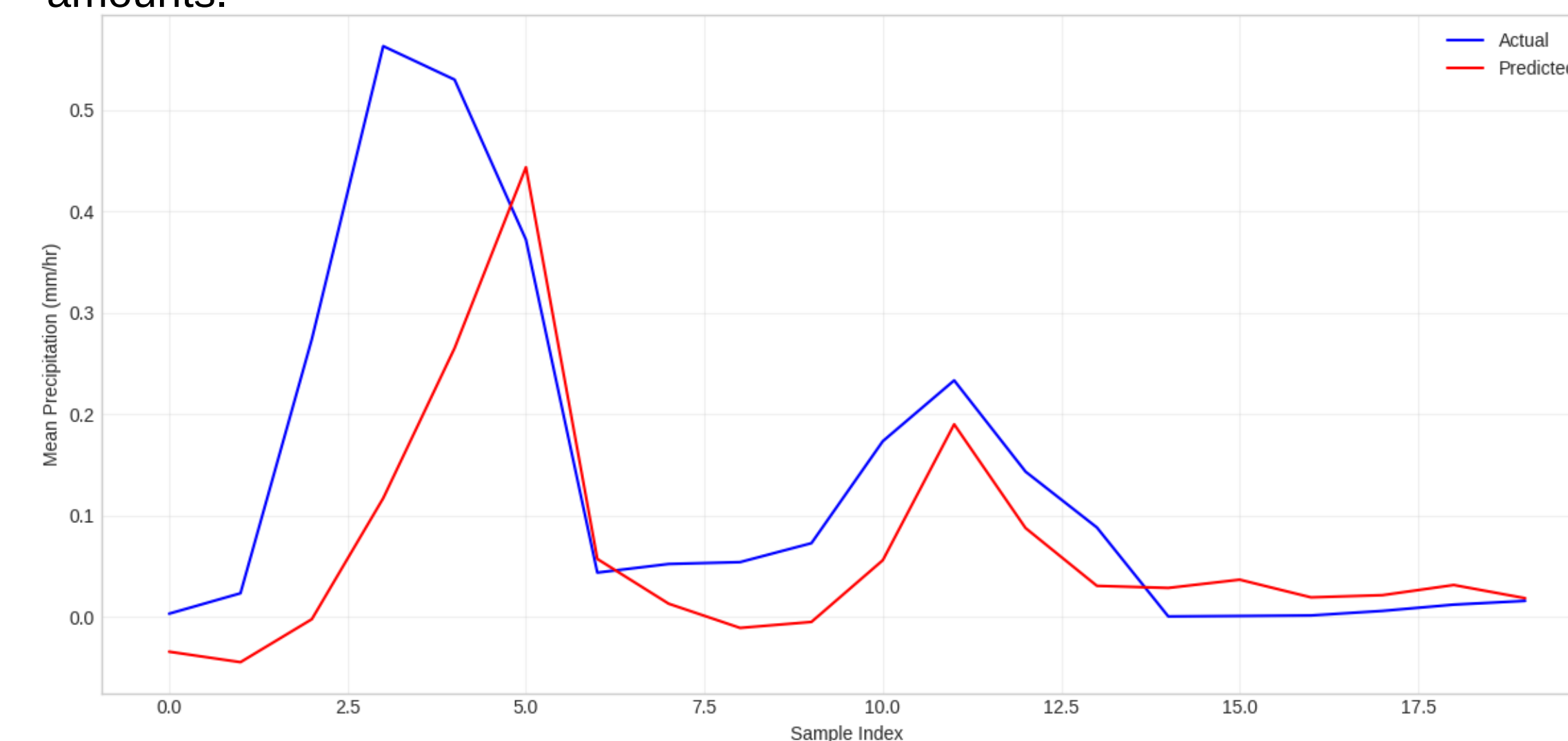


Figure 7. Time Series of Mean Precipitation for 20 Consecutive Test Samples.

- We can see the tendency to underpredict more clearly in the distribution of prediction errors in bottom-left Figure 8., with a clear left skew.
- Model metrics achieved are shown in the table to the right, notably MSE, RMSE, MAE, accuracy, and precision indicate good model performance while low recall (actual positives correctly identified) and F1 (harmony of precision and recall) scores, along with the error distribution below indicate that the class imbalance is not effectively solved through this model architecture.

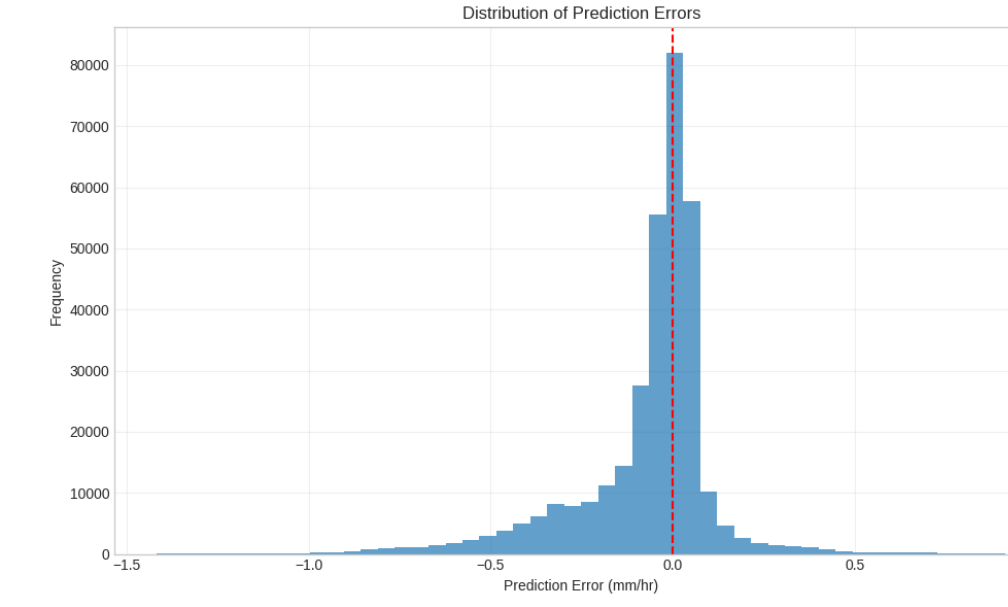


Figure 8. (Left) Distribution of Prediction Errors with clear left skew.

Figure 9. (Right) Table of Metrics used to evaluate model with test dataset.

Metric	Value
MSE	0.04
RMSE	0.20
MAE	0.12
Accuracy	0.77
Precision	0.79
Recall	0.43
F1	0.56

FUTURE WORK

Future work will involve:

- Increasing the spatial resolution of CFSV2 data to match GSMap resolution to predict precipitation with higher resolution or using higher resolution datasets.
- Investigating the disconnect in correlation between feature precipitation and labeled precipitation.
- Using dual U-net classification and regression model to mask pixels predicted as not having precipitation.
- Investigating other methods to deal with class imbalance
- Including a wider input window, multiple hourly arrays to predict precipitation at one hour some forecast hours in the future, instead of just one time (hour) as the model currently employs.
- Seeing if other CNN model architectures such as Res-Net or LSTM could be useful for taking into the temporal trends of precipitation.

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